

PHUONG (AIRI) PHAM

Los Angeles, CA | phamairi@gmail.com | LinkedIn | phuongp.com

TECHNICAL SKILLS

C++, C#, Python, HLSL, GLSL, Slang, Vulkan, DirectX 11, OpenGL, Unreal Engine, Unity, Niagara, Maya, Git, Perforce

EDUCATION

University of Southern California — B.S. Computer Science Games & M.S. Computer Science May 2025

WORK EXPERIENCE

Respawn Entertainment — Electronic Arts

Los Angeles, CA

Associate Technical Artist

Jul 2025–Present

- Developed a reusable Unreal Engine content-ingestion pipeline that validates dependencies, reorganizes paths, resolves naming and Perforce compatibility issues for 46 externally developed assets comprising 200+ files across Star Wars productions; credited on *Star Wars: Zero Company* and an upcoming *Star Wars Jedi* title.
- Automated Python-based auditing of 300+ external and 400+ internal art source files for *Apex Legends* to identify missing assets, content differences, and projected production costs.
- Scripted Maya tools to automate porting 60+ Respawn Substance Painter materials to the custom Source engine and analyze 100+ assets for performance comparison to support up-res effort.

Respawn Entertainment — Electronic Arts

Los Angeles, CA

Technical Artist Intern

May 2024–Aug 2024

- Engineered an Unreal Engine application for discovering, loading, and inspecting shared content across multiple Star Wars titles.
- Implemented Blueprint/C++ workflows for asset collection, inspection, map and mesh previews, object teleportation, and EA Shared Assets Library integration in partnership with the Technical Art Director.

Goldman Sachs

Salt Lake City, UT

Software Engineer Intern

Jun 2023–Aug 2023

- Architected an end-to-end data-ingestion service for Tax Operations that streamlined workflow to parse and validate large client datasets; stress-tested the backend with 1M+ records and 100+ MB files by benchmarking for the most memory-efficient database update strategies and utilizing multi-threaded paradigm for speed optimization.
- Designed failure handling and scalability around production edge cases—including malformed inputs, bounded error reporting, large-file transfers, and concurrent users—and built load-testing workflows to evaluate service capacity and memory behavior under stress.

GRAPHICS PROJECTS

Prime Engine Particle System & Engine Features

C++ | Demo

Oct 2024

- Built reusable C++ engine systems including a centralized physics manager, Sphere/AABB/OBB collision components, automatic component registration, collision response, gravity, and ground detection.
- Designed a component-based CPU particle framework with object-attached emitters, configurable property curves, collision behavior, transparency, normal mapping, and flipbook animation.

Interactive Ocean Simulation

Unity, HLSL, C# | Demo

Sep 2025

- Implemented a GPU-driven Unity ocean using compute shaders for Phillips-spectrum FFT waves, interactive iWave ripples, Jacobian foam, and GPU buoyancy sampling.
- Kept visual displacement and floating-object motion synchronized by sampling the same GPU-generated displacement field, avoiding duplicate CPU-side wave calculations.

GPU-Driven Interactive Grass

Unity, HLSL, C# | Demo

Sep 2024

- Developed a GPU-driven grass simulation in Unity using HLSL compute shaders, modeling procedural Bézier blades under wind, stiffness, gravity, and object collisions.
- Scaled dense grass fields through GPU compute culling and indirect rendering, with Gauss–Legendre integration maintaining consistent blade length during deformation.

GAME PROJECTS

The Veiled Ones

Los Angeles, CA

Technical Artist | Blog

Sep 2023–Aug 2024

- Profiled CPU-bound Unity HDRP scenes with the Frame Debugger, identifying shader and keyword divergence across 2,000+ materials as the primary cause of 100+ fragmented SRP batches.
- Built editor utilities to convert and standardize 3ds Max, Arnold, Autodesk Interactive, and HDRP Lit materials; configured occlusion culling with light baking to improve FPS from 15 to 50.
- Authored gameplay and cinematic VFX—including SDF particles, decals, UV-distorted fire, smoke, and full-screen effects—while reducing overdraw and preserving the game’s horror aesthetic.